Annals of Biomedical Engineering, Vol. 47, No. 10, October 2019 *(© 2018)* pp. 2094–2108
https://doi.org/10.1007/s10439-018-2088-5

Special Issue on the NCAA-DoD CARE Consortium Research

Drill-Specific Head Impacts in Collegiate Football Practice: Implications for Reducing "Friendly Fire" Exposure

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(Received 18 April 2018; accepted 29 June 2018; published online 9 July 2018)

Abstract—This study investigated drill-specific head impact biomechanics in a Division 1 collegiate football team using the Head Impact Telemetry System (HITS). A total of 32,083 impacts were recorded across 2 years of practices. Precise tracking of instrumented athletes, head impacts, and drill participation allowed quantification of hits sustained per person per minute (H/P/M) for each specific drill. We found significant H/P/M variability between 14 drills and player position, ranging from 0.02 to 0.41 H/P/M for Linemen and 0.01 to 0.15 H/P/M for Non-Linemen. Impact magnitude data are also reported for practice term (Spring, Training Camp, In-Season) and dress-type (Helmets Only, Spyders, Shells, Full Pads). Recommendations for shortening high-risk drills, based on H/P/M drill impact frequencies, suggest possible "friendly fire" reductions of 1000 impacts for Linemen and 300 impacts for Non-Linemen over their collegiate career. Over 80% of potentially avoidable head impacts were attributable to just three drills—"Team Run," "Move the Field," and "Team." Recommending drill-specific modifications based on practical considerations (the drill's impact frequency, dress-types when performing the drill, and duration) could improve acceptance from coaches and efficiently reduce head impact exposure without drastically altering overall practice structure.

Keywords—HIT System, Subconcussive, Concussion, Repetitive head impact exposure, Football practice drills, Impact frequency.

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INTRODUCTION

Concussion risk exists in many sport and non-sport activities (recreational accidents, motor vehicle accidents, falls, assault, *etc.*) and represents a broad public health issue. Millions of athletes participate in contact sports across youth, high school, collegiate, and professional levels despite an inherent risk of head impacts and sport-related concussion (SRC). However, repetitive head impact exposure (RHIE) is relatively unique to certain collision sports. SRC and RHIE are now central to medical and public interests given concerns for resultant short- and long-term cognitive, emotional, and behavioral changes.[2,21]

Prevailing theories posit that the risk for long-term health problems, while likely not uniform across all collision sport participants, may be most attributable to a combination of extensive RHIE in conjunction with multiple SRCs, rather than single or isolated instances of SRC throughout a playing career without RHIE.[2,21] Evidence also exists suggesting measurable functional and/or structural brain changes can result from single seasons of collision sport participation.[1,3,17] In American football, attempts to address such concerns initially involved in-game rule changes focused on reducing SRC risk by modifying the highest-risk plays (e.g., altering kickoff rules[30]) and penalizing dangerous technique (e.g., "targeting" and spearing, protecting "defenseless players"). Practice sessions are another appealing avenue for risk reduction given the high frequency of head impacts from teammate collisions (i.e., "friendly fire") and the greater ability to control player activity (i.e., drill type) during practice compared to game scenarios.[6,10,14]

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More recently, the National Collegiate Athletic Association (NCAA) has systematically restricted practice times and eliminated two-a-days in hopes of ultimately reducing RHIE and injury risk.[18,23] Some institutions have gone so far as to implement helmetless practices, with promising results of reduced RHIE.[35] These efforts to reduce RHIE may be enhanced by more targeted approaches to practice modifications that consider drill-specific impact risks rather than non-specific recommendations such as, for example, practice time restriction.

Previous biomechanics research indicated the majority of RHIE occurs during practices, but impact rates and SRC risk are greatest in games.[8,14,34] Recent studies in youth football have characterized drill-specific impact frequency and magnitude differences, and have shown significant variability within a given practice. As one might expect, "tackling drills" were associated with higher frequency of head impacts though highest impact frequency drills were not always the same as the drills that had the highest impact magnitudes.[12,26] These findings highlight the possibility of identifying particular drills with the greatest risk of "friendly fire" and, subsequently, modifying practice schedules to lessen time spent performing these drills. This would optimize RHIE risk reduction.

Detailed characterization of drill-specific head impact biomechanics has not occurred at the collegiate level. This is perhaps due to, at least in part, the complexity of practice structures involving multiple drill stations and more position-specific drills than may be expected at youth or high school levels. Existing evidence in college athletes suggests an overall high frequency of RHIE,[15] and that SRC risk is higher for games than practices.[19] However, practice term (i.e., spring vs. training camp vs. in-season practices[24]) and dress-type (i.e., helmets only vs. shells vs. full pads[31]) are likely important moderators of these relationships.

Taken together, there appear to be ample opportunities for efficiently reducing "friendly fire" impacts. The primary aim of the present study was to characterize drill-specific impact frequency for a Division 1 collegiate football team based on the number of head impacts sustained per person for each minute they performed a given drill. We hypothesized that certain drills would be readily identified as "higher risk" as demonstrated by other researchers' previous work in youth football. Estimated reductions to RHIE could then be calculated based on both non-specific and targeted recommendations for shortening drill times. Secondary aims included exploration of impact magnitude differences between drill, practice terms, and practice dress type. These data may allow practical modifications to existing practice structures for mitigation of "friendly fire" head impacts.

METHODS

Participants

Forty-seven NCAA Division 1 collegiate football players from the University of Florida (mean ± SD age at start of enrollment, 19.0 ± 1.0 years) participated in this study. A total of 169 practice sessions were studied over a 2-year period. This encompassed two complete football activity years, each composed of three distinct practice terms: two 4-week spring practice camps ("Spring," 2016 and 2017), two 4-week fall training camps ("Training Camp," 2016 and 2017), and two competitive seasons ("In-Season," 2016–2017 and 2017–2018) spanning ~ 18 and ~ 14 weeks, respectively. Twelve individuals participated in 1 term, 9 participated in 2 terms, 6 participated in 3 terms, 9 participated in 4 terms, 5 participated in 5 terms, and 6 participated in all 6 practice terms. Study inclusion required that the athlete wear the helmet sensors for at least one practice session, regardless of whether they sustained a measureable impact; though, all participants contributed head impact data in our sample (i.e., no instrumented players recorded zero impacts across their participation period). Players were stratified into two positional groups for most analyses based on previously described differences in head impact biomechanics[8,28]: Linemen (offensive and defensive), and Non-Linemen (linebackers, defensive backs, running backs, wide receivers, tight ends). Table 1 describes the number of participants by position for each of the six terms. All athletes signed informed consent documents approved by the University's Institutional Review Board prior to having their helmets fitted with impact accelerometers.

Head Impact Telemetry System (HITS) Instrumentation

Participating athletes wore helmets equipped with the HITS (Simbex, Lebanon, NH). Data are obtained from six spring-mounted single-axis accelerometers, which were embedded within Riddell Speed, Speed Classic, and Speed Flex helmets. The signal transducer was linked to a laptop computer in the sideline response system (SRS) through radiowave transmission (903–927 MHz). The information was stored on an onboard memory system (up to 100 impacts) or was immediately transferred to the laptop computer system (8-bit, 1000 Hz/channel). The downloaded impacts were processed through a validation algorithm that excludes extraneous impacts (e.g., dropping a helmet). All hits deemed valid by the Simbex validation algorithm were downloaded; peak head linear and rotational accelerations additionally were used for analyses

TABLE 1. Number of players per position wearing Head Impact Telemetry System (HITS) sensors for at least one practice session during each season.

	2016			2017		
Position	Spring	Training camp	In-season	Spring	Training camp	In-season
Linemen	13	11	14	14	13	12
Offensive	9	8	10	10	9	9
Defensive	4	3	4	4	4	3
Non-linemen	13	15	17	9	6	7
Linebacker	7	4	4	3	2	3
Defensive back	3	5	5	3	2	2
Running back	2	2	4	1	0	0
Wide receiver	1	3	3	2	2	2
Tight end	0	1	1	0	0	0
Overall	26	26	31	23	19	19

involving impact magnitude. Previous studies have validated the HITS against laboratory models,[4,27] though some have raised concern about its limitations.[20,25]

Data Collection

We focused exclusively on head impacts sustained during practice events and did not include game impact data given the presumed relatively equal participation across athletes wearing HIT sensors during practice, and the specific aim of characterizing drill-specific impact frequency. We defined these as "friendly fire" impacts based on their occurrence resulting directly from athletic engagement with one's own teammates, regardless of intentionality, and their occurrence during practice where there is possibility for greater activity modifications.

No impacts were associated with a clinically diagnosed concussion. A dedicated research assistant was present for all practice sessions throughout the study period to ensure proper sensor functioning, document player participation, and record drill times. Practice scripts were obtained for each practice session and cross-referenced with HITS data to organize drill-specific impact data. For every practice session, the name of each drill and the amount of time spent performing the drill (minutes) were tracked, while closely accounting for any deviations from the script as well as positional drill differences throughout the practice (e.g., linemen occasionally performing different drills than non-linemen positions). Additionally, the number of athletes wearing HIT sensors each day was recorded so that investigators knew exactly how many athletes accounted for the number of impacts sustained during a given drill.

Head impact data were exported from the HIT SRS and uploaded into SPSS v24.0[33] (Armonk, NY), which was used for performing all analyses in the study. All impact data from both practices and games were initially exported before excluding game impacts. Only impacts registering a linear acceleration greater than 10 *g* were included in analyses. Impacts are linked to a unique player identification number and include the date and time of each impact. This allowed the research assistant to cross-reference each recorded impact with the drill during which it occurred.

Calculating Head Impact Frequency

Based on the number of recorded impacts for a drill, the number of athletes wearing HIT sensors while the drill was being performed, and the total number of minutes spent performing the drill, the hits/person/minute of drill time (H/P/M) was calculated for all 14 different drills (see Table 2). All practice-related activities were accounted for by these 14 drill classifications. "Scrimmages" were specifically designated as Drill #1 due to the unique structure of intra-squad scrimmages relative to the other drills, which were randomly assigned a number from 2 to 14. If drills were performed more than once in the same practice, their minutes were summed. For the majority of analyses, practice terms were combined across the 2-year study period; for example, Training Camp data consisted of impacts from Training Camp 2016 and Training Camp 2017, and In-Season data consisted of In-Season 2016–2017 and In-Season 2017–2018.

Drill names and general practice structure remained consistent over the course of the study period. Of note, minute-by-minute practice tracking began with Training Camp 2016 and therefore does not include Spring 2016, so only Spring 2017 data was available for calculating "Spring" H/P/M data. Spring 2016 impacts were included in analyses that did not involve H/P/M data.

TABLE 2. Description of each of the 14 practice drills.

Drill codes	Drill names	Drill description
#1	Scrimmage	11 vs. 11 live intra-squad scrimmages between the offense and defense
#2	Team run	11 vs. 11 full team period where each rep was a scripted run play only
#3	Move the field	11 vs. 11 full team period in which the offense played against the defense while the ball was spotted at different points on the field throughout the drill
#4	Indy	Individual position groups separated and practiced specific techniques germane to their position group; this period was highly variable between position groups and specific techniques were at the discretion of the position coach
#5	Teach	Offense and defense practiced new plays, formations, and schemes at a slower pace than a "Team" period. Often 11 vs. 11, but at times 7 vs. 7 or 11 vs. 0 ("on air" reps)
#6	Team	Full team period of 11 vs. 11 drills that consisted of scripted offensive plays with full contact
#7	Blitz	11 vs. 11 full team period that consisted of defensive blitz plays
#8	Fastballs	Fast-paced team period of short duration, in which the offense was given a small number of plays and/or a short amount of time to score from midfield; often performed 11 vs. 11 but at times an 11 vs. 0 setting
#9	Special teams	Punting, kickoffs, kick returns, field goals, and special teams circuits; primarily Non-Linemen participation
#10	Goal line	11 vs. 11 full team period that took place within the 20-yard line; plays were often very close to the goal line
#11	2-minute	11 vs. 11 full team period that simulated a game situation where the offense had less than 2 min to drive down the field for either a field goal or a touchdown
#12	3rd Down	11 vs. 11 full team period that simulated a 3rd down situation
#13	Blitz/O-D walk	Walkthrough period that occurred before stretching; sometimes performed as 11 vs. 11 and sometimes as 11 vs. 0
#14	Skelly	7 vs. 7 period that focused primarily on skill positions and receiver routes, typically without Linemen participating. When Linemen participated, it was usually at half-speed and only included partial line groups that made the drill 9 vs. 9 rather than 7 vs. 7

We also provided a list and description of the 14 drills to experienced members of the football team's athletic training staff and equipment staff present for the majority of practice sessions over the 2-year study period. They were each asked to independently rank-order the "top 5 worst drills for head impacts per player per minute" of drill time, with "1" representing the drill they believed was associated with the highest head impact frequency. They were not provided with any impact data prior completing their lists. Descriptive analyses were performed to characterize the accuracy of their predictions and determine medical staff's and equipment staff's abilities to spot higher-risk drills.

Calculating Potential Head Impact Reduction Estimates

We calculated two potential but arbitrary methods for reducing exposure to head impacts during the 14 practice drills over a 4-year collegiate football career. To estimate the number of times a drill would be performed over the course of a 4-year collegiate career, the known number of times completing each of the 14 drills throughout the 2 years' worth of Training Camp and In-Season data were each multiplied by 2, and the Spring 2017 data was multiplied by 4. Note that one of the 14 "drills" represents intra-squad scrimmages, which are typically completed separate from the other drill types and performed for 1–2 h at a time. Our study period included two intra-squad scrimmages per year during both Spring (distinct from the "Spring Game") and Training Camp practices. Therefore, it is expected that 16 total scrimmages would occur in the course of a 4-year career. The first method for reducing exposure quantified the estimated reduction associated with performing each of the 14 drills for 1 min less, which equates to 13 fewer total minutes per practice, and 1 min less per intra-squad scrimmage. The second method was a more targeted reduction based on identifying drills that (a) have high H/P/M, and (b) are performed for relatively greater number of minutes. In this scenario, 15 min of time deductions were selectively distributed over the highest H/P/M common practice drills, and an additional 10 min were deducted from intra-squad scrimmages. Separate head impact reduction estimates were calculated for Linemen and Non-Linemen.

Exploring Magnitude Differences by Drill, Season, and Practice Dress-Type

We additionally investigated how overall and drill-specific H/P/M and impact magnitudes might differ by both practice term (Spring vs. Training Camp vs. In-Season) and practice dress-type (Helmets Only, Spyders, Shells, Full Pads—Practice, and Full

Pads—Scrimmage; see eTable A Electronic Supplementary Material). Sample size limitations precluded the ability to stratify impact data by drill, practice term, and practice dress type simultaneously. These exploratory analyses were therefore limited to drill-specific differences between practice terms, and drill-specific differences between dress types. Nonparametric analyses (Kruskal–Wallis with *post hoc* Mann–Whitney *U*) were performed when comparing impact magnitudes (both linear and rotational) between drills and practice terms. Atypically high magnitude impacts were determined using a sample-derived cutoff for linear and rotational accelerations greater than the 95th percentile and analyzed using χ^2 tests.

Effect size estimates for nonparametric analyses were calculated using the common language effect size statistic proposed by Grissom and Kim,[22] also known as a probability index. This is interpreted similarly to an area under the curve (AUC) statistic; it is the likelihood that a value randomly selected from group A will be greater than a value randomly selected from group B ($p_{a>b}$). Therefore, a $p_{a>b}$ of 0.5 means there is an equal likelihood that the value from group A will be higher or lower than the value of group B (i.e., random chance or negligible effect). The following cutoffs are used in this study based on previously published relationships between AUC values and traditional conventions for Cohen's *d* effect sizes[32]: 0.56 (equivalent to $d = 0.2$) is a small effect, 0.64 (equivalent to $d = 0.5$) is a medium effect, and 0.71 (equivalent to $d = 0.8$) is a large effect.

RESULTS

A total of 32,083 "friendly fire" impacts were recorded across the Spring, Training Camp, and In-Season practice terms for the 2016 and 2017 football seasons, of which 23,580 (73.5%) were accounted for by Linemen. The grand mean (median) linear and rotational accelerations were 28.5 *g* (23.5 *g*) and 1792.0 rad s^{-2} (1563.2 rad s^{-2}), respectively. Excluding Spring 2016 where minute-to-minute drill tracking was unavailable, 28,389 total impacts were reported. For Linemen, 21,140 impacts were recorded during 14 drills totaling 20,225 min of drill time, with an average of 9.0 Linemen wearing HIT sensors per drill. For Non-Linemen, 7249 impacts were recorded during the 14 drills totaling 14,361 min of drill time, with an average of 8.7 Non-Linemen wearing HIT sensors per drill.

Drill-Specific H/P/M

Tables 3 and 4 show the total recorded impacts for each drill during each of the five practice terms when minute-to-minute data was available for Linemen and Non-Linemen, respectively. There is clear variability in H/P/M across the different drills for the overall practice data, ranging from 0.02 (Drill #9, "Special Teams") to 0.41 (Drill #2, "Team Run") H/P/M for Linemen, and from 0.01 (Drill #13, "Blitz/O-D Walk") to 0.15 (Drill #2, "Team Run") H/P/M for Non-Linemen. For all drills except "Special Teams", H/P/M was the same or noticeably higher for Linemen than Non-Linemen.

A total of 10 members of the football team's athletic training staff ($N = 6$, range of college football experience = 3–57 years) and equipment staff ($N = 4$, range of college football experience = 6–13 years) completed surveys rank-ordering estimates of which five drills they believed had the highest H/P/M. For Linemen, 10/10 raters ranked the highest H/P/M drill ("Team Run") among their top 5, and 7/10 ranked it in the top 2. Conversely, 0/10 raters identified the 2nd highest H/P/M drill ("Fastballs") among their top 5, while 3/10, 8/10, and 8/10 ranked the 3rd ("Move the Field"), 4th ("Scrimmage"), and 5th ("Blitz") highest H/P/M drills among their top 5. None of the five lowest H/P/M drills were consistently identified in any rater's rank-order top 5 predictions.

For Non-Linemen, 8/10 raters ranked the highest H/P/M drill ("Team Run") among their top 5, and 3/10 ranked it in the top 2. Eight of 10 raters ranked the 2nd highest H/P/M drill ("Scrimmage") among their top 5, while 4/10, 6/10, and 0/10 ranked the 3rd ("Move the Field"), 4th ("Blitz"), and 5th ("Fastballs") highest H/P/M drills among their top 5. Once again, none of the five lowest H/P/M drills were consistently identified in any rater's rank-order top 5 predictions for Non-Linemen.

Impact Reduction Estimations

Estimated reduction to head impact exposure was calculated for a 4-year collegiate football career for Linemen and Non-Linemen if each drill were shortened by 1 min each time it was performed (see eTables B and C in Electronic Supplementary Material). In this scenario, each typical practice would have to be shortened by 13 min, while each intra-squad scrimmage (Drill #1) would be shortened by 1 min. Collegiate-career exposure would be reduced by approximately 480 impacts for each lineman and 175 impacts for each non-lineman using this method.

A more targeted approach to time reductions was then made based on examination of the H/P/M data and the available number of minutes from which time could be reasonably deducted. Tables 5 and 6 describe the estimated reduction to head impact exposure for participants in the Linemen and Non-Linemen groups,

TABLE 3. Head impacts (HIs) sustained per sensor-wearing participant per minute of drill time (hits/person/minute; H/P/M).

Registered head impacts (> 10 g) sustained per person for each minute of drill time

Linemen (DL, OL)

	Overall						Spring ('17)				Training camp ('16 and '17)				In-season ('16–'17 and '17–'18)			
					95% CI													
Drill No.	# HI	Avg. *N*	Total Min.	H/P/M	Lower	Upper	# HI	Avg. *N*	Total Min.	H/P/M	# HI	Avg. *N*	Total Min.	H/P/M	# HI	Avg. *N*	Total Min.	H/P/M
1	1396	9.8	610	**0.23**	**0.22**	**0.24**	475	10.0	175	**0.27**	921	9.7	435	**0.22**	–	–	–	–
2	4355	8.9	1199	**0.41**	**0.41**	**0.42**	585	9.5	113	**0.54**	1207	8.2	305	**0.49**	2563	9.4	781	**0.35**
3	1950	8.8	800	**0.28**	**0.27**	**0.29**	395	9.5	116	**0.36**	681	8.3	270	**0.30**	874	9.3	414	**0.23**
4	4808	8.9	9340	**0.06**	**0.06**	**0.06**	787	9.7	969	**0.08**	1355	8.4	3388	**0.05**	2666	9.4	4983	**0.06**
5	943	8.9	960	**0.11**	**0.10**	**0.11**	117	9.7	76	**0.16**	119	8.0	240	**0.06**	707	9.4	644	**0.12**
6	2515	9.0	1415	**0.20**	**0.19**	**0.20**	194	9.6	72	**0.28**	331	7.7	137	**0.31**	1990	9.4	1206	**0.18**
7	790	8.0	446	**0.22**	**0.20**	**0.23**	39	9.7	19	**0.21**	115	7.7	65	**0.23**	636	9.5	362	**0.18**
8	589	8.8	211	**0.32**	**0.28**	**0.33**	94	9.7	18	**0.54**	251	8.1	76	**0.41**	244	9.4	117	**0.22**
9	263	8.9	1934	**0.02**	**0.01**	**0.02**	24	9.7	107	**0.02**	53	8.2	430	**0.01**	186	9.4	1397	**0.01**
10	1313	9.4	675	**0.21**	**0.20**	**0.21**	58	10.5	15	**0.37**	131	8.9	71	**0.21**	1124	9.4	589	**0.20**
11	661	8.9	524	**0.14**	**0.13**	**0.15**	129	9.8	80	**0.16**	260	8.1	168	**0.19**	272	9.2	276	**0.11**
12	638	9.1	410	**0.17**	**0.15**	**0.18**	42	8.0	12	**0.44**	71	9.6	35	**0.21**	525	9.4	363	**0.15**
13	668	8.9	1536	**0.05**	**0.04**	**0.05**	141	9.7	149	**0.10**	260	8.2	451	**0.07**	267	9.4	936	**0.03**
14	251	9.3	165	**0.16**	**0.12**	**0.17**	65	10.3	41	**0.15**	49	8.3	26	**0.23**	137	9.2	98	**0.15**
Total	21,140	9.0	20225	**0.12**	**0.11**	**0.12**	3145	9.7	1962	**0.17**	5804	8.4	6097	**0.11**	12191	9.4	12166	**0.11**

The total number of head impacts captured during a drill (# HI) was divided by the average number of sensor-wearing participants for that drill (Avg. *N*). This value was then divided by the total minutes spent performing the drill (Total Min.) to calculate the H/P/M for that drill.

Avg. N average number of sensor-wearing participants, *CI* confidence interval, *DL* defensive linemen, *HI* head impacts, *H/P/M* hits per person per minute of drill time, *Min.* minutes, *No.* number, *OL* offensive linemen.

TABLE 4. Head impacts (HIs) sustained per sensor-wearing participant per minute of drill time (hits/person/minute; H/P/M).

Registered head impacts (> 10 *g*) sustained per person for each minute of drill time

Non-linemen (LB, DB, RB, WR, TE)

	Overall						Spring ('17)				Training camp ('16–'17)				In-season ('16–'17 and '17–'18)			
					95% CI													
Drill No.	# HI	Avg. *N*	Total Min.	H/P/M	Lower	Upper	# HI	Avg. *N*	Total Min.	H/P/M	# HI	Avg. *N*	Total Min.	H/P/M	# HI	Avg. *N*	Total Min.	H/P/M
1	483	7.3	610	**0.11**	**0.09**	**0.12**	139	7.5	175	**0.11**	343	7.5	435	**0.11**	–	–	–	**--**
2	1586	8.8	1199	**0.15**	**0.15**	**0.16**	119	8.7	113	**0.12**	402	8.5	305	**0.15**	1065	9.4	781	**0.15**
3	585	8.7	800	**0.08**	**0.07**	**0.09**	86	8.8	116	**0.09**	246	9.1	370	**0.07**	253	9.4	414	**0.07**
4	989	8.7	1975	**0.06**	**0.05**	**0.06**	131	8.8	309	**0.05**	293	8.7	764	**0.04**	565	9.3	902	**0.07**
5	458	8.8	960	**0.05**	**0.05**	**0.06**	39	8.8	76	**0.06**	86	8.6	240	**0.04**	333	9.5	644	**0.05**
6	661	8.5	1415	**0.05**	**0.05**	**0.06**	33	8.7	72	**0.05**	72	7.0	137	**0.08**	556	9.4	1206	**0.05**
7	234	8.0	446	**0.07**	**0.05**	**0.07**	12	8.8	19	**0.15**	63	7.9	65	**0.12**	159	9.8	362	**0.04**
8	127	8.7	211	**0.07**	**0.07**	**0.07**	2	8.8	18	**0.01**	49	8.3	76	**0.08**	76	9.6	117	**0.07**
9	724	8.8	1934	**0.04**	**0.04**	**0.05**	43	8.8	107	**0.05**	124	8.6	430	**0.03**	557	9.3	1397	**0.04**
10	334	9.5	675	**0.05**	**0.04**	**0.06**	9	9.5	15	**0.06**	33	9.7	71	**0.05**	292	9.6	589	**0.05**
11	230	8.8	524	**0.05**	**0.04**	**0.05**	27	9.1	80	**0.04**	126	8.1	168	**0.09**	77	9.4	276	**0.03**
12	142	9.5	410	**0.04**	**0.03**	**0.04**	2	8.0	12	**0.02**	26	11.8	35	**0.06**	114	9.2	363	**0.03**
13	113	8.8	1536	**0.01**	**0.01**	**0.01**	18	8.8	149	**0.01**	39	8.6	451	**0.01**	56	9.4	936	**0.01**
14	583	8.7	1666	**0.04**	**0.04**	**0.04**	64	8.6	122	**0.06**	185	8.5	396	**0.05**	334	9.3	1148	**0.03**
Total	7249	8.7	14,361	**0.06**	**0.05**	**0.06**	724	8.7	1383	**0.06**	2087	8.6	3943	**0.07**	4438	9.43	9135	**0.05**

The total number of head impacts captured during a drill (# HI) was divided by the average number of sensor-wearing participants for that drill (Avg. *N*). This value was then divided by the total minutes spent performing the drill (Total Min.) to calculate the H/P/M for that drill. H/P/M statistics and corresponding confidence interval values are rounded to the nearest hundredth. *Avg. N* average number of sensor-wearing participants, *CI* confidence interval, *DB* defensive back, *HI* head impacts, *H/P/M* hits per person per minute of drill time, *LB* linebacker, *Min.* minutes, *No.* number, *RB* running back, *TE* tight end, *WR* wide receiver.

respectively. In this scenario, each typical practice would have to be shortened by 15 min, while each intra-squad scrimmage (Drill #1) would be shortened by 10 min. Collegiate-career exposure would be reduced by almost 1000 impacts for each linemen and 300 impacts for each on-lineman using this method.

Taken together, "Team Run", "Move the Field", and "Team" were among the highest risk, based primarily on H/P/M data, the amount of time spent in the drill, and relative consistency across positional groups. Considering these factors offers practical advantages for generalized practice modifications. "Fastballs" was also considered high risk, but this drill is performed infrequently and for short durations (typically less than 3 min), resulting in fewer total associated impacts and limiting feasible modifications. Of the approximately 990 head impacts that could be avoided by Linemen with a targeted practice time deduction approach (Table 5), around 790 (~ 80%) would be deducted from these three drills alone. Of the approximately 300 head impacts that could be avoided by Non-Linemen (Table 6), around 250 (~ 83%) would be deducted from these 3 drills.

Clear differences in H/P/M between Linemen and Non-Linemen and across drills were observed (see eFigure A in Electronic Supplementary Material). While the overall H/P/M rate for Linemen is two times greater than that of Non-Linemen, this difference is amplified to approximately three to four times greater in the higher-risk drills.

Drill-Specific Impact Magnitudes

The proportion of sustained impacts above the sample-derived cutoff for the 95th percentile impact magnitude for linear (62.3 g) and rotational acceleration (3820.8 rad s^{-2}) was calculated for each individual drill (see eTables D and E in Electronic Supplementary Material). We also performed categorical comparisons between the three "high risk" drills and all other drills. Overall, players were at greater risk of sustaining a high linear acceleration impact (> 95th percentile) during a "high risk" drill (combined rate for "Team Run," "Move the Field," and "Team" drills = 5.6%) than during "all other drills" (combined rate = 4.6%, χ^2 = 17.641,

TABLE 5. Estimated reduction to Linemen head impact exposure over a 4-year collegiate football career if high risk drills targeted for length reduction each time drill is performed.

Estimated 4-year impact reduction with targeted time deductions

Linemen (DL, OL)

	Spring				Training camp				In-season				Collegiate career total			
Drill No.	H/P/M	Est. # Sess.*	Min. Red.	HI Red.	H/P/M	Est. # Sess.*	Min. Red.	HI Red.	H/P/M	Est. # Sess.*	Min. Red.	HI Red.	H/P/M	Est. # Sess.	Min. Red.	HI Red.
1	0.27	8	10	**21.6**	0.22	8	10	**17.6**	–	0	–	–	0.23	16	10	**36.8**
2	0.54	44	4	**95.0**	0.49	56	4	**109.8**	0.35	130	4	**182.0**	0.41	230	4	**377.2**
3	0.36	44	3	**47.5**	0.30	58	3	**52.2**	0.23	136	3	**93.8**	0.28	238	3	**199.9**
4	0.08	48	0	**0.0**	0.05	70	0	**0.0**	0.06	206	0	**0.0**	0.06	324	0	**0.0**
5	0.16	48	0	**0.0**	0.06	68	0	**0.0**	0.12	198	0	**0.0**	0.11	314	0	**0.0**
6	0.28	32	4	**35.8**	0.31	36	4	**44.6**	0.18	200	4	**144.0**	0.20	268	4	**214.4**
7	0.21	8	1	**1.7**	0.23	18	1	**4.1**	0.18	128	1	**23.0**	0.22	154	1	**33.9**
8	0.54	48	1	**25.9**	0.41	52	1	**21.3**	0.22	104	1	**22.9**	0.32	204	1	**65.3**
9	0.02	48	0	**0.0**	0.01	70	0	**0.0**	0.01	206	0	**0.0**	0.02	324	0	**0.0**
10	0.37	8	1	**3.0**	0.21	20	1	**4.2**	0.20	136	1	**27.2**	0.21	164	1	**34.4**
11	0.16	28	1	**4.5**	0.19	46	1	**8.7**	0.11	68	1	**7.5**	0.14	142	1	**19.9**
12	0.44	4	0	**0.0**	0.21	10	0	**0.0**	0.15	102	0	**0.0**	0.17	116	0	**0.0**
13	0.10	48	0	**0.0**	0.07	68	0	**0.0**	0.03	206	0	**0.0**	0.05	322	0	**0.0**
14	0.15	12	0	**0.0**	0.23	8	0	**0.0**	0.15	38	0	**0.0**	0.16	58	0	**0.0**
Total HI reduction	235				263				500				≈ *990*			

Intra-squad scrimmages (Drill #1) last much longer than a typical practice drill, thus allowing for greater feasible reduction. The combined time reduction for other drills (#2–#14) equates to 15 fewer practice minutes per day in this sample, and a potential reduction of almost 1000 "friendly fire" head impacts over a collegiate practice career.

*Estimated collegiate career total sessions for Spring Camp calculated by multiplying known number of Spring 2017 sessions by 4, and known number of Fall Camp (2016 and 2017) and Fall Season (2016 and 2017) sessions by 2.

DL defensive linemen, *Est. # Sess.* estimated number of sessions, *HI Red.* head impact reduction, *H/P/M* head impacts per person per minute of drill time, *Min. Red.* proposed number of minutes by which a drill would be reduced, *No.* number, *OL* offensive linemen, *Yr.* year.

TABLE 6. Estimated reduction to Non-Linemen head impact exposure over a 4-year collegiate football career if high risk drills targeted for length reduction each time drill is performed.

Estimated 4-year impact reduction with targeted time deductions

Non-Linemen (LB, DB, RB, WR, TE)

	Spring				Training camp				In-season				Collegiate career total			
Drill No.	H/P/M	Est. # Sess.*	Min. Red.	HI Red.	H/P/M	Est. # Sess.*	Min. Red.	HI Red.	H/P/M	Est. # Sess.*	Min. Red.	HI Red.	H/P/M	Est. # Sess.*	Min. Red.	HI Red.
1	0.11	8	10	**8.8**	0.11	8	10	**8.8**	–	–	–	–	0.10	16	10	**16.0**
2	0.12	44	4	**21.1**	0.15	56	4	**33.6**	0.15	130	4	**78.0**	0.15	230	4	**138.0**
3	0.09	44	3	**11.9**	0.07	58	3	**12.2**	0.07	136	3	**28.6**	0.08	238	3	**57.1**
4	0.05	48	0	**0.0**	0.04	70	0	**0.0**	0.07	206	0	**0.0**	0.06	324	0	**0.0**
5	0.06	48	0	**0.0**	0.04	68	0	**0.0**	0.05	198	0	**0.0**	0.05	314	0	**0.0**
6	0.05	32	4	**6.4**	0.08	36	4	**11.5**	0.05	200	4	**40.0**	0.05	268	4	**53.6**
7	0.15	8	1	**1.2**	0.12	18	1	**2.2**	0.04	128	1	**5.1**	0.06	154	1	**9.2**
8	0.01	48	1	**0.5**	0.08	52	1	**4.2**	0.07	104	1	**7.3**	0.07	204	1	**14.3**
9	0.05	48	0	**0.0**	0.03	70	0	**0.0**	0.04	206	0	**0.0**	0.04	324	0	**0.0**
10	0.06	8	1	**0.5**	0.05	20	1	**1.0**	0.05	136	1	**6.8**	0.05	164	1	**8.2**
11	0.04	28	1	**1.1**	0.09	46	1	**4.1**	0.03	68	1	**2.0**	0.05	142	1	**7.1**
12	0.02	4	0	**0.0**	0.06	10	0	**0.0**	0.03	102	0	**0.0**	0.03	116	0	**0.0**
13	0.01	48	0	**0.0**	0.01	68	0	**0.0**	0.01	206	0	**0.0**	0.01	322	0	**0.0**
14	0.06	36	0	**0.0**	0.05	64	0	**0.0**	0.03	208	0	**0.0**	0.04	308	0	**0.0**
Total HI reduction	52				78				168				≈300			

Intra-squad scrimmages (Drill #1) last much longer than a typical practice drill, thus allowing for greater feasible reduction. The combined time reduction for other drills (#2–#14) equates to 15 fewer practice minutes per day in this sample, and a potential reduction of approximately 300 "friendly fire" head impacts over a collegiate practice career.

DB defensive back, *Est. # Sess.* estimated number of sessions, *HI Red.* head impact reduction, *H/P/M* head impacts per person per minute of drill time, *LB* linebacker, *Min. Red.* proposed number of minutes by which a drill would be reduced, *No.* number, *RB* running back, *TE* tight end, *WR* wide receiver, *Yr.* year.

*Estimated collegiate career total sessions for Spring Camp calculated by multiplying known number of Spring 2017 sessions by 4, and known number of Fall Camp (2016 and 2017) and Fall Season (2016 and 2017) sessions by 2.

$p < 0.001$, small effect size). Results were similar for high rotational acceleration impacts (5.5% of hits in "high risk" drills vs. 4.6% in all other drills, $\chi^2 = 13.150$, $p < 0.001$, small effect size). The figure shows these and position-specific associations between "high risk" drills and high magnitude impacts for both linear (Figure A) and rotational (Figure B) acceleration (Figure 1).

Examination of the individual drill data for Linemen shows that "Scrimmage" (5.7%), "Team Run" (6.7%), and "Special Teams" (7.3%) have greater than the expected frequency of high linear acceleration impacts (i.e., more than 5.0% of impacts greater than the sample 95th percentile of 62.3 *g*), while "Scrimmage" (6.4%), "Team Run" (5.8%), and "Skelly" (7.4%) carry greater than expected risk of high rotational acceleration impacts. For Non-Linemen, "Scrimmage" (6.9%), "Team Run" (6.8%), and "Blitz" (6.3%) have greater than expected frequency of high linear acceleration impacts, while "Scrimmage" (8.7%), "Team Run" (6.7%), "Move the Field" (7.3%), "Indy" (6.3%), "Teach" (6.2%), "Blitz" (7.6%), and "Blitz/O-D Walk" (6.1%) all carry greater than expected risk of high rotational acceleration impacts.

Practice Term and Dress-Type Magnitude Differences

We examined whether overall impact magnitude differed between practice terms: Spring, Training Camp, and In-Season practices. For Linemen, there was a significant main effect of practice term for both linear (Spring > Training Camp = In-Season; $\chi^2 = 14.897$, $p = 0.001$) and rotational acceleration (Spring > Training Camp > In-Season; $\chi^2 = 92.578$, $p < 0.001$). For Non-Linemen, there was also a significant main effect of practice term for both linear (Spring > Training Camp = In-Season; $\chi^2 = 11.408$, $p = 0.003$) and rotational acceleration (Spring = Training Camp > In-Season; $\chi^2 = 32.827$, $p < 0.001$). Statistically significant *post hoc* comparisons with effect size estimates of linear and rotational acceleration differences are illustrated in eFigure B. The effect sizes suggest that observed differences were clinically negligible. The categorical association

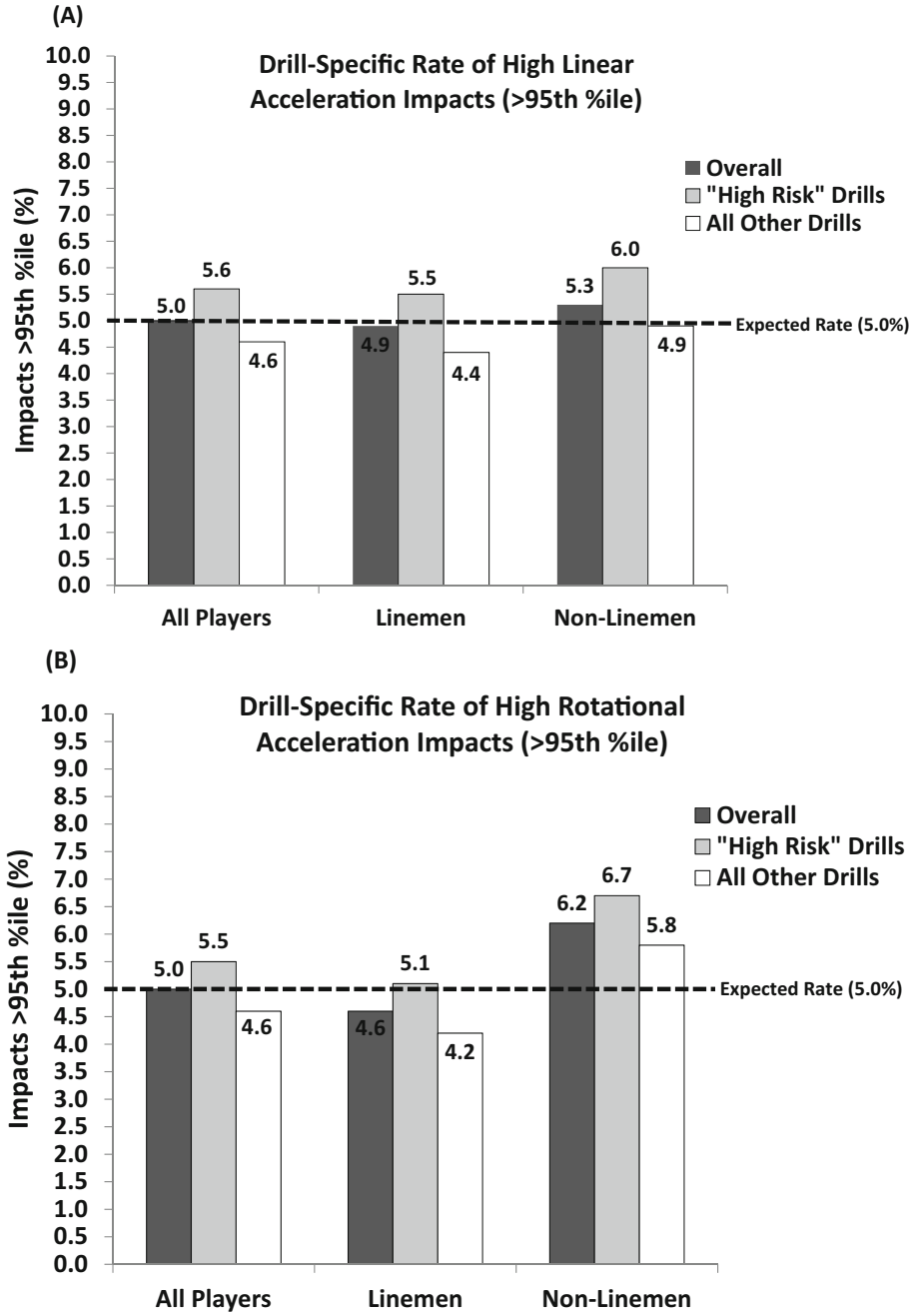

FIGURE 1. (a) and (b) Combined percentage of high magnitude impacts (> 95th percentile) sustained overall, during three "high risk drills" ("Team Run," "Move the Field," and "Team"), and during "all other drills" (11 other drills). Rates are presented for both high linear (a) and high rotational (b) acceleration impacts, with further stratification by position group (Linemen vs. Non-Linemen). Values above 5.0% indicate greater than expected frequency of high magnitude impacts. χ^2 analyses were performed comparing rates between "high risk" drills and "all other drills" (light grey compared to white columns). Results indicated significantly greater risk of high magnitude impacts (both linear and rotational) during the three "high risk" drills compared to all other drills for All Players, Linemen, and Non-Linemen ($p < 0.05$, small effect sizes), with the exception of Non-Linemen rotational impacts ($p = 0.075$). Effect sizes were small for all differences.

between practice term and rates of impacts > 95th percentile magnitude was significant for both Linemen and Non-Linemen ($p < 0.01$ for all). The greatest risk of high magnitude impacts (both linear and rotational) tended to occur during Spring practice (small effect sizes; see eFigure C).

Lastly, we compared impact magnitude between the different practice dress types (see eTable A in Elec-

tronic Supplementary Material for dress-type description) using the overall sample of recorded impacts (combined across all drills) since not every drill is performed with every dress type. For Linemen, there was a significant main effect of dress type for both linear ($\chi^2 = 248.655$, $p < 0.001$) and rotational acceleration ($\chi^2 = 257.044$, $p < 0.001$). For Non-Linemen, there was also a significant main effect of dress type for both linear ($\chi^2 = 48.667$, $p < 0.001$) and rotational acceleration ($\chi^2 = 45.649$, $p < 0.001$). Statistically significant *post hoc* comparisons with effect size estimates for linear and rotational magnitude differences are illustrated in eFigure D, most of which were between a small and medium effect. The categorical association between dress-type and rates of impacts > 95th percentile magnitude was significant for both Linemen and Non-Linemen ($p < 0.01$ for all). The greatest risk of high magnitude impacts (both linear and rotational) tended to occur with Full Pads-Practice and Full Pads-Scrimmage, and the lowest risks occurred with Spyders and Helmets Only dress-types (small effect sizes; see eFigure E).

DISCUSSION

The primary goal of this study was to characterize drill-specific impact frequency in a Division 1 collegiate football program. In so doing, we identified frequency variability across the 14 drills performed and proposed targeted time reductions that could potentially reduce "friendly fire" collisions by around 300 impacts for each non-lineman and 1000 impacts for each lineman in their collegiate careers. This would effectively eliminate the equivalent of a full year of impact exposure based on published average impact frequency estimates.[29] Three drills in particular, "Team Run", "Move the Field", and "Team", account for over 80% of these impacts. When combined, these three "high risk" drills were also associated with more frequent high magnitude (greater than 95th percentile) impacts compared to all other drills.

Comparing to Youth Football Data

This study uniquely describes drill-specific differences in impact biomechanics at the collegiate level and also proposes a feasible means for significantly reducing "friendly fire" impact exposure throughout a 4-year college football career. Kelley and colleagues report similar data from a youth football sample and noted impact frequencies as high as 0.59 hits per person per minute (H/P/M) of drill time,[26] which is higher than the present sample's highest rate for Linemen (0.41 H/P/M during "Team Run"), and much higher than the highest rate for Non-Linemen (0.15 H/P/M for "Team Run"). Campolettano *et al.* also report a higher hits per minute rate for select drills (95% CI 0.42–1.13),[12] but do not specifically indicate the number of drill participants. In addition to the present study being specific to collegiate athletes, we were able to separate Linemen from Non-Linemen impact rates. Our H/P/M data were based on over 28,000 practice impacts sustained by 47 total participants, compared to just 2125[26] and 6183[12] impacts in the other studies. Lower numbers of available impacts per drill in these previous studies may increase error surrounding impact rate estimates. There may also be differences in general practice structure across levels of play. Drill descriptions from these previous studies suggest individual (e.g., 1 vs. 1) or small group drills are more common at the youth level where there is greater focus on skill development than in college where practice tends to emphasize game-specific scenarios (often full-sided 11 vs. 11).

Of note, frequency definitions vary across studies and include season impacts per person (at times stratified by practice or game),[8,11,15,28,34] impacts per hour,[12] and impacts per person per minute,[26] often making direct comparisons difficult. However, consistent findings further supported by the present study are: (1) there are position-specific differences in impact exposure (Linemen with greatest exposure) and (2) impact frequency itself varies greatly within the different elements of a football season (practice term differences, drill-specific differences, *etc.*). This suggests that gross estimates of impacts per season likely mask important person-level and situation-level exposure differences that could more meaningfully inform modification recommendations. For example, recommending a general time restriction for practices, while certain to reduce overall RHIE,[6,10] may not be nearly as effective as more targeted restrictions on time spent performing specific drills. It is worth considering that there may be unintended consequences of further reducing the time spent developing skills and techniques in practice, such as potentially increasing risk for musculoskeletal or head injury in game situations. This concern warrants further research.

Regardless, preliminary supportive evidence for RHIE reduction through mitigating "friendly fire" impacts exists at the youth and high school levels. Cobb *et al.* studied three different youth football teams and found significantly lower impact frequency during practices for athletes on the team that enforced strict limits on contact practice participation (6.2 impacts per practice) compared to the other two teams (12.9 and 9.5 impacts per practice).[13] Broglio and colleagues speculated that limiting high school contact practices to 1/week would result in 18% fewer head impacts

overall, and that complete elimination of contact practices would reduce exposure by 39%.[6] Empirical support was then provided in 2013 when a statewide policy in Michigan restricted allowable collision practices to 2/week starting in 2014. This policy was associated with an average exposure reduction of 42%, despite no measurable change in impact magnitudes.[10]

Our data suggest that upwards of 1000 and 300 impacts may be avoided for each individual "Linemen" and "Non-Linemen," respectively, with *targeted*, drill-specific reductions of 15 min per practice session and 10 min per intra-squad scrimmage throughout a college career. This is consistent with previous work at the high school level concluding that Linemen stand to benefit most from practice modifications.[6,10] Taken together, further reducing RHIE appears feasible at all levels of American football; however, additional research must investigate whether such reductions ultimately leads to a clinically relevant decline in SRC rate or risk for long-term health problems.

Considerations for Impact Magnitude and Dress-Type

Our study primarily focused on impact frequency irrespective of magnitude based on an increased emphasis on reducing overall RHIE rather than just SRC risk *per se*, though both are undoubtedly important prevention targets. Minor magnitude differences were observed across drills and practice terms, but the most relevant findings came from the investigation of dress-type differences. Surprisingly, dress type minimally influenced linear and rotational magnitudes. Other research investigating dress-type differences is mixed.

Reynolds *et al.* described dress-type impact differences based on both frequency (helmets only < shells < full pads) and magnitude (helmets only < shells = full pads), highlighting possible inter-practice variability.[31] Mihalik and colleagues found no difference between Helmets Only and Full Pads practices.[28] At face value, this is both concerning and somewhat confusing given the presumed differences in practice intensity and degree of contact between these dress types. More in-depth study of this phenomenon is certainly warranted. However, it is possible, if not likely, that gross comparisons of mean or median magnitude values masks the potentially greater risk for high magnitude impacts associated with certain dress-types.

Since the vast majority of head impacts fall below 30 g, a categorical comparison of the highest risk impacts may be more appropriate. In support of this, our data showed that players were two to three times more likely to sustain impacts greater than the sample-specific 95th percentile (~ 62 g linear, ~ 3800 rad s^{-2} rotational) during full pads practices compared to other dress types (eFigure E). The influence of dress-type on risk of high magnitude impacts appeared particularly strong for Non-Linemen. Therefore, we believe that dress-type (i.e., practice intensity and degree of contact) likely does contribute to risk for higher magnitude impacts, but the differences are not immediately apparent based on mean or median acceleration differences.[31] Since some drills are performed in multiple dress types, we speculate that further focusing time reductions in these drills when wearing, for example, Full Pads as opposed to Spyders dress may convey even greater risk reduction.

Practical Considerations for Implementing Modifications

Modifying practice structure presents practicality and feasibility hurdles. From a scientific standpoint, reasonable minds may disagree on whether to focus mitigation efforts on general RHIE vs. risk for high severity impacts (i.e., risk for SRC-causing impacts). However, it is worth noting that SRC susceptibility has been shown to be influenced not only by impact magnitude but also impact density, or an individual sustaining multiple impacts within a short amount of time.[5] Therefore, structuring practice by separating high risk drills or dividing them into multiple segments with lower-risk drills interspersed could be helpful. Drill-specific data show that drills with the highest impact frequency do not necessarily also present the greatest risk for high magnitude impacts.[26] Lastly, there is the reality that coaching staffs must buy into proposed recommendations. Some drills, given their short duration, do not lend themselves to modifications unless they are just outright removed, which coaches may resist.

For example, the present study identified a potentially high risk drill ("Fastballs") as a candidate for time reductions; however, this drill is only performed sporadically throughout the year and typically lasts for 3 min or less, leaving little room for adjusting duration. Campolettano similarly identified a "King of the Circle" drill considered high risk but performed for the shortest amount of time among all drills.[12] A balance of feasibility and compliance may be best achieved by targeting drills considered to be among the highest risk, that are also performed consistently, and of sufficient duration to reduce minutes without requiring outright elimination. Another modification strategy might involve a redistribution of the time deducted from higher risk drills to more skill-based, technical drills that carry lower risk of head impacts. While this may result in overall slightly less RHIE reduction, total face-to-face coaching time would be preserved while also efficiently eliminating many "friendly fire" impacts.

Limitations

A key strength of our study was the use of a daily onsite research assistant affording us the ability to track minute-to-minute impacts throughout often complex practice structures, while accounting for positional (Linemen vs. Non-Linemen) variability in drills. Daily tracking of instrumented participants was also essential because the drills performed vary from practice to practice, and every consented athlete did not participate in every practice session across the 2-year study period for various reasons (e.g., injury, graduation, removing sensors due to discomfort). Therefore, we were confident in knowing which specific athletes accounted for the impacts recorded during a given practice session and, by extension, the drills performed for that session.

However, the monumental efforts of our research assistant (ZB) are also a potential limitation since this level of oversight may not be feasible at other institutions even if they have head impact monitoring equipment. Along those lines, these data are specific to one Division 1 institution and its practices. While similar drills are likely performed at other universities, the exact H/P/M impact frequency data and calculated head impact reduction values may not directly translate. They are instead meant to serve as representing the possible outcomes associated with drill-specific impact tracking and targeted time reductions. The H/P/M data for "Scrimmages" may underestimate true impact frequency since participation disproportionately favors starters or "first team" athletes, and the number of actual reps per instrumented player was unknown. We suspect, but cannot guarantee, that participation in other drills was more evenly distributed among all instrumented athletes regardless of depth chart position.

Possible instrumental limitations are measurement error within the HITS[4] and the 100 ms reset window after a recorded impact which could result in missed impacts, potentially affecting magnitude and frequency estimates, respectively. Additionally, both practice structure and overall head impact biomechanics are expected to differ between collegiate, high school, and youth football participation,[8,9,15,16,26] limiting the generalizability of these data to other levels of play. While 47 total participants accounted for the 32,000 + impacts in our study, numbers of participants and impacts varied across season and practice term, and we did not verify these impacts *via* video recording. We also had to extrapolate data from these two seasons when estimating 4-year exposure totals, which assumes that the two study seasons are representative of a full college career of practices, and that the athletes wearing helmet sensors are representative of all rostered athletes. Lastly, player positions were not equally represented in our sample, and larger data collection efforts[7] may be able to investigate position-specific differences in college on a more granular level than Linemen and Non-Linemen.

Drill-specific impact tracking at an institutional level would specifically inform targeted recommendations for practice modifications, provided the institution possesses some form of head impact monitoring system like the HITS and at least one dedicated staff member. However, this is often financially unrealistic on a wide scale, particularly at high school and youth participation levels. In the absence of a head impact monitoring system, the H/P/M data from the present study (Tables 3 and 4) may be used in conjunction with the drill descriptions (Table 2) for determining the relative impact frequencies for similarly structured drills. Additionally, our survey of experienced members of the athletic training staff and equipment staff suggest that highest H/P/M drills may be readily identifiable without a head impact monitoring system, and lower H/P/M drills are rarely misperceived as having a high impact frequency.

CONCLUSIONS

Impact frequency varies between practice drills in collegiate football. Shortening the duration of specific drills may significantly reduce RHIE resulting from "friendly fire" collisions and could be more effective than non-specific recommendations for limiting contact practice time. Linemen, in particular, would benefit from drill-specific time modifications. Recommending drill-specific modifications based on practical considerations (the drill's impact frequency, dress-types when performing the drill, and duration) could result in buy-in from coaches and efficiently reduce head impact exposure without drastically altering overall practice structure.

ELECTRONIC SUPPLEMENTARY MATERIAL

The online version of this article (https://doi.org/10.1007/s10439-018-2088-5) contains supplementary material, which is available to authorized users.

ACKNOWLEDGMENTS

We wish to thank the University of Florida's University Athletic Association for their continued support of concussion research at the institution. We also thank the student-athletes who participated in this study.

CONFLICT OF INTEREST

The HIT System was purchased with funding provided in part by the Florida High Tech Corridor Council (FHTCC) and Banyan Biomarkers, Inc., with support from the NFL-GE Head Health Challenge. The authors declare no conflicts of interest pertaining to this study. BA received funding support from Banyan Biomarkers, Inc. for efforts related to statistical analysis and manuscript preparation. JC received support through the NCAA-DOD CARE Consortium as site PI (Office of the Assistant Secretary of Defense for Health Affairs through the Psychological Health and Traumatic Brain Injury Program under Award NO W81XWH-14-2- 0151), Banyan Biomarkers, Inc., and the Florida High Tech Corridor.

REFERENCES

[1]Bahrami, N., D. Sharma, S. Rosenthal, E. M. Davenport, J. E. Urban, B. Wagner, Y. Jung, C. G. Vaughan, G. A. Gioia, J. D. Stitzel, and C. T. Whitlow. Subconcussive head impact exposure and white matter tract changes over a single season of youth football. *Radiology* 281:919–926, 2016.

[2]Baugh, C. M., J. M. Stamm, D. O. Riley, B. E. Gavett, M. E. Shenton, A. Lin, C. J. Nowinski, R. C. Cantu, A. C. McKee, and R. A. Stern. Chronic traumatic encephalopathy: neurodegeneration following repetitive concussive and subconcussive brain trauma. *Brain Imaging Behav*. 6:244–254, 2012.

[3]Bazarian, J. J., T. Zhu, J. Zhong, D. Janigro, E. Rozen, A. Roberts, T. Zhu, J. Zhong, D. Janigro, E. Rozen, A. Roberts, H. Javien, K. Merchant-Borna, B. Abar, and E. G. Blackman. Persistent, long-term cerebral white matter changes after sports-related repetitive head impacts. *PLoS ONE* 9:e94734, 2014.

[4]Beckwith, J. G., R. M. Greenwald, and J. J. Chu. Measuring head kinematics in football: correlation between the head impact telemetry system and Hybrid III headform. *Ann. Biomed. Eng*. 40:237–248, 2012.

[5]Broglio, S. P., A. Lapointe, K. L. O'Connor, and M. McCrea. Head impact density: a model to explain the elusive concussion threshold. *J. Neurotrauma* 34:2675–2683, 2017.

[6]Broglio, S. P., D. Martini, L. Kasper, J. T. Eckner, and J. S. Kutcher. Estimation of head impact exposure in high school football: implications for regulating contact practices. *Am. J. Sports Med*. 41:2877–2884, 2013.

[7]Broglio, S. P., M. McCrea, T. McAllister, J. Harezlak, B. Katz, D. Hack, B. Hainline, and C. C. Investigators. A national study on the effects of concussion in collegiate athletes and US Military Service Academy Members: The NCAA–DoD Concussion Assessment, Research and Education (CARE) Consortium structure and methods. *Sports Med*. 47:1437–1451, 2017.

[8]Broglio, S. P., J. J. Sosnoff, S. Shin, X. He, C. Alcaraz, and J. Zimmerman. Head impacts during high school football: a biomechanical assessment. *J. Athl. Train*. 44:342–349, 2009.

[9]Broglio, S. P., T. Surma, and J. A. Ashton-Miller. High school and collegiate football athlete concussions: a biomechanical review. *Ann. Biomed. Eng*. 40:37–46, 2012.

[10]Broglio, S. P., R. M. Williams, K. L. O'Connor, and J. Goldstick. Football players' head-impact exposure after limiting of full-contact practices. *J. Athl. Train*. 51:511–518, 2016.

[11]Brolinson, P. G., S. Manoogian, D. McNeely, M. Goforth, R. Greenwald, and S. Duma. Analysis of linear head accelerations from collegiate football impacts. *Curr. Sports Med. Rep*. 5:23–28, 2006.

[12]Campolettano, E. T., S. Rowson, and S. M. Duma. Drill-specific head impact exposure in youth football practice. *J. Neurosurg. Pediatr*. 18:536–541, 2016.

[13]Cobb, B. R., J. E. Urban, E. M. Davenport, S. Rowson, S. M. Duma, J. A. Maldjian, C. T. Whitlow, A. K. Powers, and J. D. Stitzel. Head impact exposure in youth football: elementary school ages 9–12 years and the effect of practice structure. *Ann. Biomed. Eng*. 41:2463–2473, 2013.

[14]Crisco, J. J., R. Fiore, J. G. Beckwith, J. J. Chu, P. G. Brolinson, S. Duma, T. W. McAllister, A.-C. Duhaime, and R. M. Greenwald. Frequency and location of head impact exposures in individual collegiate football players. *J. Athl. Train*. 45:549–559, 2010.

[15]Crisco, J. J., B. J. Wilcox, J. G. Beckwith, J. J. Chu, A.-C. Duhaime, S. Rowson, S. M. Duma, A. C. Maerlender, T. W. McAllister, and R. M. Greenwald. Head impact exposure in collegiate football players. *J. Biomech*. 44:2673–2678, 2011.

[16]Daniel, R. W., S. Rowson, and S. M. Duma. Head impact exposure in youth football. *Ann. Biomed. Eng*. 40:976–981, 2012.

[17]Davenport, E. M., C. T. Whitlow, J. E. Urban, M. A. Espeland, Y. Jung, D. A. Rosenbaum, C. T. Whitlow, J. E. Urban, M. A. Espeland, Y. Jung, D. A. Rosenbaum, G. A. Gioia, A. K. Powers, J. D. Stitzel, and J. A. Maldjian. Abnormal white matter integrity related to head impact exposure in a season of high school varsity football. *J. Neurotrauma* 31:1617–1624, 2014.

[18]Division I 20/8-Hour Rule Materials. www.ncaa.org. National Collegiate Athletic Association (NCAA). 29 March. http://www.ncaa.org/division-i-20/8-hour-rule-materials.

[19]Dompier, T. P., Z. Y. Kerr, S. W. Marshall, B. Hainline, E. M. Snook, R. Hayden, and J. E. Simon. Incidence of concussion during practice and games in youth, high school, and collegiate American football players. *JAMA Pediatr*. 169:659–665, 2015.

[20]Duma, S. M., and S. Rowson. RE: on the accuracy of the Head Impact Telemetry (HIT) system used in football helmets. *J. Biomech*. 47:1557–1558, 2014.

[21]Gavett, B. E., R. A. Stern, and A. C. McKee. Chronic traumatic encephalopathy: a potential late effect of sport-related concussive and subconcussive head trauma. *Clin. Sports Med*. 30:179–188, 2011.

[22]Grissom, R. J., and J. J. Kim. Effect Sizes for Research: Univariate and Multivariate Applications. New York: Routledge, pp. 177–202, 2012.

[23]Hosick, M. DI Council Votes to Eliminate Football Two-a-Days. www.ncaa.org. National Collegiate Athletic Association (NCAA). 29 March. http://www.ncaa.org/about/resources/media-center/news/di-council-votes-eliminate-football-two-days.

[24]Houck, Z., B. Asken, R. Bauer, J. Pothast, C. Michaudet, and J. Clugston. Epidemiology of sport-related concussion

in an NCAA Division I football bowl subdivision sample. *Am. J. Sports Med.* 44:2269–2275, 2016.

[25]Jadischke, R., D. C. Viano, N. Dau, A. I. King, and J. McCarthy. On the accuracy of the Head Impact Telemetry (HIT) System used in football helmets. *J. Biomech.* 46:2310–2315, 2013.

[26]Kelley, M. E., J. M. Kane, M. A. Espeland, L. E. Miller, A. K. Powers, J. D. Stitzel, and J. E. Urban. Head impact exposure measured in a single youth football team during practice drills. *J. Neurosurg. Pediatr.* 20:489–497, 2017.

[27]Manoogian, S., D. McNeely, M. Goforth, G. Brolinson, and S. Duma. Head acceleration is less than 10 percent of helmet acceleration during a football impact. *Biomed. Sci. Instrum.* 42:383–388, 2006.

[28]Mihalik, J. P., D. R. Bell, S. W. Marshall, and K. M. Guskiewicz. Measurement of head impacts in collegiate football players: an investigation of positional and event-type differences. *Neurosurgery* 61:1229–1235, 2007.

[29]Montenigro, P. H., M. L. Alosco, B. M. Martin, D. H. Daneshvar, J. Mez, C. E. Chaisson, C. J. Nowinski, R. Au, A. C. McKee, and R. C. Cantu. Cumulative head impact exposure predicts later-life depression, apathy, executive dysfunction, and cognitive impairment in former high school and college football players. *J. Neurotrauma* 34:328–340, 2017.

[30]Ocwieja, K. E., J. P. Mihalik, S. W. Marshall, J. D. Schmidt, S. C. Trulock, and K. M. Guskiewicz. The effect of play type and collision closing distance on head impact biomechanics. *Ann. Biomed. Eng.* 40:90–96, 2012.

[31]Reynolds, B. B., J. Patrie, E. J. Henry, H. P. Goodkin, D. K. Broshek, M. Wintermark, and T. J. Druzgal. Practice type effects on head impact in collegiate football. *J. Neurosurg.* 124:501–510, 2016.

[32]Rice, M. E., and G. T. Harris. Comparing effect sizes in follow-up studies: ROC Area, Cohen's d, and r. *Law Hum. Behav.* 29:615, 2005.

[33]SPSS, I. SPSS Statistical Software. Armonk, NY: IBM Corporation, 2013.

[34]Schnebel, B., J. T. Gwin, S. Anderson, and R. Gatlin. In vivo study of head impacts in football: a comparison of National Collegiate Athletic Association Division I versus high school impacts. *Neurosurgery* 60:490–496, 2007.

[35]Swartz, E. E., S. P. Broglio, S. B. Cook, R. C. Cantu, M. S. Ferrara, K. M. Guskiewicz, and J. L. Myers. Early results of a helmetless-tackling intervention to decrease head impacts in football players. *J. Athl. Train.* 50:1219–1222, 2015.